

Exam Content List - Terraform Associate 004

6min | Terraform



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This guide lists the specific exam objectives and resources you can study to demonstrate those objectives. Review our [learning path](#) for a start-to-finish list of resources you can use to prepare for the exam. While the tutorials listed may use specific cloud providers to introduce the learning objectives, provider-specific knowledge is not necessary for the exam.

Visit the [HashiCorp Cloud Engineer Certification page](#) for more information and to register for the exam.

Content differences between the 003 and 004 exams

If you took the Terraform Associate (003) exam version, review the following information to prepare for the new version.

Exam updates summary:

- Four new topics
 - 4f: depends_on and create_before_destroy lifecycle rules
 - 4g: Validate configuration using custom conditions

- 4h: Ephemeral values and write-only arguments
- 8c: Describe how to organize and use HCP Terraform workspaces and projects
- Tests on Terraform version 1.12
- Includes HCP Terraform content

Exam content list

 HashiCorp Developer

 Terraform

1	Infrastructure as Code (IaC) with Terraform	Documentation	Tutorials
1a	Explain what IaC is	What is Terraform? Infrastructure as Code in a Private or Public Cloud	Introduction to Infrastructure as Code
1b	Describe the advantages of IaC patterns	Why Terraform Infrastructure as Code in a Private or Public Cloud	Introduction to Infrastructure as Code
1c	Explain how Terraform manages multi-cloud, hybrid cloud, and service-agnostic workflows	Multi-Cloud Deployment	Introduction to Infrastructure as Code
2	Terraform fundamentals	Documentation	Tutorials
2a	Install and version Terraform providers	Providers Specifying Provider Requirements Dependency Lock File Provider Block Reference	Lock and Upgrade Provider Versions Initialize Terraform Configuration
2b	Describe how Terraform uses providers	Provider Block Reference Provider Overview Provider Requirements How Terraform Works with Plugins	Initialize Terraform Configuration

2c	Write Terraform configuration using multiple providers	Provider Block Reference Provider Requirements	Define Infrastructure with Terraform Resources
2d	Explain how Terraform uses and manages state	Purpose of Terraform State	Manage Resources in Terraform State
3	Core Terraform workflow	Documentation	Tutorials
3a	Describe the Terraform workflow	Core Terraform Workflow	Create Infrastructure Create a Terraform Plan Apply Terraform Configuration Initialize Terraform Configuration
3b	Initialize a Terraform working directory	Dependency Lock File Command: init	Initialize Terraform Configuration
3c	Validate a Terraform configuration	Command: validate	Validate your Configuration
3d	Generate and review an execution plan for Terraform	Resource Graph Command: plan	Create a Terraform Plan
3e	Apply changes to infrastructure with Terraform	Command: apply	Apply Terraform Configuration
3f	Destroy Terraform-managed infrastructure	Command: destroy	Destroy Infrastructure
3g	Apply formatting and style adjustments to a configuration	Command: fmt	Format the Configuration
4	Terraform configuration	Documentation	Tutorials

4a	Use and differentiate <code>resource</code> and <code>data</code> blocks	Resources Data Sources	Query Data Sources
4b	Refer to resource attributes and create cross-resource references	Resource Addressing References to Named Values	Create Resource Dependencies
4c	Use variables and outputs	Manage values in modules Output Block Reference Variable Block Reference	Customize Terraform Configuration with Variables Output Data from Terraform
4d	Understand and use complex types	Complex Types	Customize Terraform Configuration with Variables
4e	Write dynamic configuration using expressions and functions	Built-in Functions	Create Dynamic Expressions Perform Dynamic Operations with Functions
4f	Define resource dependencies in configuration	Resource Graph	Create Resource Dependencies
4g	Validate configuration using custom conditions	Validate Your Configuration	Use Checks to Validate Infrastructure Validate Modules with Custom Conditions
4h	Understand best practices for managing sensitive data, including secrets management with Vault	Manage Sensitive Data Vault Provider for Terraform	Upgrade RDS Major Version Protect Sensitive Input Variables

			Inject Secrets into Terraform Using the Vault Provider
5	Terraform modules	Documentation	Tutorials
5a	Explain how Terraform sources modules	Find and Use Modules	Modules Overview Use Registry Modules in Configuration
5b	Describe variable scope within modules	Module Block Reference Manage values in modules	Build and Use a Local Module
5c	Use modules in configuration	Module Composition Output Block Reference Manage values in modules	Use Registry Modules in Configuration
5d	Manage module versions	Module Block Reference: Version Argument	Use Registry Modules in Configuration
6	Terraform state management	Documentation	Tutorials
6a	Describe the local backend	Backend Block Configuration Backend Type: local	Initialize Terraform Configuration
6b	Describe state locking	State Storage and Locking State Locking	
6c	Configure remote state using the backend block	State Storage and Locking	

		Backend Block Configuration	
6d	Manage resource drift and Terraform state	Terraform State Refactor Terraform State Refresh-Only Mode Removed Block Reference Moved Block Reference	Manage Resource Drift Use Refresh-Only Mode
7	Maintain infrastructure with Terraform	Documentation	Tutorials
7a	Import existing infrastructure into your Terraform workspace	Command: import Import Existing Resources	Import Terraform Configuration
7b	Use the CLI to inspect state	State Command	Manage Resources in State
7c	Describe when and how to use verbose logging	Debugging Terraform	Enable Terraform Logging
8	HCP Terraform	Documentation	Tutorials
8a	Use HCP Terraform to create infrastructure	Workspaces HCP Terraform Workflow HCP Terraform Overview Remote Operations	HCP Terraform Get Started Collection
8b	Describe HCP Terraform collaboration and governance features	Explorer HCP Private Registry Overview	Authenticate Providers with Dynamic Credentials

		Change Requests Overview HCP Terraform Policy Enforcement Projects Overview Health HCP Terraform Teams	Drift Detection Variable Sets Detect Drift and Enforce Policies Enforce OPA Policies
8c	Describe how to organize and use HCP Terraform workspaces and projects	Run Triggers Variable Sets Projects Overview	Connect Workspaces with Run Triggers Organize Workspaces with Projects
8d	Configure and use HCP Terraform integration	Use CLI with HCP Terraform Connect to HCP Terraform Migrate Terraform State to HCP Terraform Remote Operations Command: login	Migrate State to HCP Terraform Upgrade Terraform Version in HCP Terraform Collaborate Using HCP Terraform

Continue studying

Review the [learning.path](#) to practice all of the exam objectives. Check out the [sample questions](#) to review the exam question format.

[Sign up for the exam here!](#)

[Previous](#)[Associate learning path \(004\)](#)[Next](#)[Associate sample questions \(004\)](#)

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